

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	V Deepa Dharshini	Reg. No.:	192525213
Section / Batch:	B.Tech AIML(2 nd year)	Date:	18-08-2026

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues, in-depth analysis, Innovative, feasible solutions with strong justification

Question Type	Description	Questions	Total Marks
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1 – Q2	100

SECTION A – CASE STUDY

**SIMATS Online Programming Lab**

V Deepa Dharshini
192525213RD

CO3-AT3-Case Study

← Back

Language: Python C C++ Java

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

2/2 answered

Q1 Divide and Conquer 50 marks

Case Study: Engineering Simulation System (Matrix Multiplication)
An engineering simulation tool performs repeated matrix operations for structural analysis. Analyze how divide and conquer matrix multiplication improves computational efficiency. Identify challenges in large-scale matrix handling. Apply optimized methods such as Strassen's algorithm. Analyze performance and design a solution, justifying its advantages.

Your Code

```
1 def add_matrix(A, B):  
2     n = len(A)  
3     return [[A[i][j] + B[i][j] for j in range(n)] for i in range(n)]  
4  
5  
6 def subtract_matrix(A, B):  
7     n = len(A)  
8     return [[A[i][j] - B[i][j] for j in range(n)] for i in range(n)]  
9  
10  
11 def strassen(A, B):  
12     n = len(A)  
13  
14     if n == 1:  
15         return [A[0][0] + B[0][0]]
```

Input

2
1 2
3 4
5 6
...

Output

Enter matrix size: Enter
Matrix A:
Enter Matrix B:
Resultant Matrix:
13 22

CO3-AT3-Case Study

← Back

Language: Python C C++ Java

QUESTIONS
Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

2/2 answered

Q2 Divide and Conquer

50 marks

Case Study: Banking Transaction Processing (Merge Sort)

A banking system processes and sorts millions of daily transactions based on timestamps for auditing and reporting. Analyze how Merge Sort can be applied to ensure stable and efficient sorting. Identify key issues such as large data volume and need for stability. Apply divide and conquer concepts to explain the approach. Analyze time complexity and performance. Design a suitable solution and justify its effectiveness in this scenario.

Your Code

```
1 def merge_sort(transactions):
2
3     if len(transactions) <= 1:
4         return transactions
5     mid = len(transactions) // 2
6     left = transactions[:mid]
7     right = transactions[mid:]
8     left = merge_sort(left)
9     right = merge_sort(right)
10    return merge(left, right)
11
12 def merge(left, right):
13     result = []
14     i = 0
```

Input

```
5
26-08-18-14:30 T103 5000
2026-08-18-09:15 T101 2500
2026-08-18-12:45 T102 3000
```

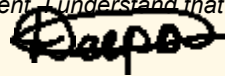
Output

```
Transactions sorted by
timestamp:
2026-08-18-09:15 T101 2500.0
2026-08-18-11:10 T104 1500.0
2026-08-18-12:45 T102 3000.0
2026-08-18-16:20 T105 7000.0
```

Code of Conduct

I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:


RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Mark Evaluation:

Case Study	Understanding of case (25)	Applications of concepts (25)	Analysis (20)	Solution Design (20)	Clarity (10)	Total Marks (Each - 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Signature: _____	Signature: _____
Signature: _____		
Date: _____	Date: _____	Date: _____